

1.

$$x_1, \dots, x_n \mid \mu \sim N(\mu, \sigma^2)$$

$$\mu \sim N(\mu_0, \tau_0^2)$$

$$\bar{x} = \frac{1}{n} \sum x_i$$

(a) $\mu \mid x \sim N(\mu_n, \tau_n^2)$

$$\tau_n^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2} \right)^{-1},$$

$$\mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2} \right)$$

derive posterior precision

$$L(\mu|x) \propto \exp\left[-\frac{1}{2\sigma^2}\sum(x_i-\mu)^2\right]$$

$$\Pi(\mu) \propto \exp\left[-\frac{1}{2\sigma_0^2}(\mu-\mu_0)^2\right]$$

$$= \Pi(\mu|x) \propto \exp\left[-\frac{1}{2}\left(\frac{\sum(x_i-\mu)^2}{\sigma^2}+\frac{(\mu-\mu_0)^2}{\sigma_0^2}\right)\right]$$

$$\sum (x_i - \mu)^2 = \sum x_i^2 - 2n\bar{x}\mu + n\mu^2$$

$$(\mu - \mu_0)^2 = \mu^2 - 2\mu_0\mu + \mu_0^2$$

$$\pi(\mu|x) \propto \exp\left(-\frac{1}{2}\left(\frac{n}{\sigma^2} + \frac{1}{\tau_0^2}\right)\mu^2 - 2\left(\frac{n\bar{x}}{\sigma^2} + \frac{\mu_0}{\tau_0^2}\right)\mu\right)$$

$$\exp\left\{-\frac{1}{2}\rho_0^2\left(\kappa^2-2\kappa\kappa_n\right)\right\}$$

$$\Rightarrow \sqrt{\frac{1}{r^2}=\frac{1}{\sigma^2}+\frac{h}{r_0^2}}\qquad \kappa_n=r_n^2\left(\frac{\kappa_0}{r_0^2}+\frac{n\bar{x}}{\sigma^2}\right)$$

(6) $\mu_n = w\bar{x} + (1-w)\mu_0 \quad w \in [0,1]$

$$\mu_n = \frac{\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2}}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}} \Rightarrow \frac{n/\sigma^2}{1/\tau_0^2 + n/\sigma^2}\bar{x} + \frac{1/\tau_0^2}{\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}}\mu_0$$

$$\boxed{\mu_n = w\bar{x} + (1-w)\mu_0}$$

$$\boxed{w = \frac{n/\sigma^2}{1/\tau_0^2 + n/\sigma^2} = \frac{n\tau_0^2}{\sigma^2 + n\tau_0^2}}$$

the weight w is our confidence / total spread

(b) If $\frac{n}{\sigma^2} > \frac{1}{\tau_0^2}$, then μ is closer to $\bar{x}$ and posterior means closer to $\bar{x}$

If prior precision $\frac{1}{\tau_0^2}$ is, then μ is closer to 0 and our posterior mean is closer to μ_0

(c)

$$x_f \mid \mu \sim N(\mu, \sigma^2) \text{ and } \mu \mid x \sim N(\mu_n, \tau_n^2)$$

$$p(x_f \mid x) = \int p(x_f \mid \mu)\, \pi(\mu \mid x)\, d\mu$$

$$x_f = \mu + \underline{\varepsilon_f}$$

$$E[X_{ij}|x] = E[E(X_{ij}|\mu_i, x)|x] = E[\mu_i|x] = \mu_n$$

$$Var(X_{ij}|x) = E[Var(X_{ij}|\mu_i, x)|x] +$$

$$Var(E[X_{ij}|\mu_i, x]|x)$$

$$= \sigma^2 + \tau_n^2$$

$$\boxed{X_{ij}|x \sim N(\mu_n, \sigma^2 + \tau_n^2)}$$

(d)

$n = 16, \quad \bar{x} = 5.4, \quad \sigma^2 = 4, \quad \mu \sim N(4, 1)$

$\mu_0 = 4, \quad \tau_0^2 = 1$

find posterior mean, variance, predictive dist

variance:

$$\tau_1^2 = \left(\frac{1}{\tau_0^2} + \frac{n}{\sigma^2}\right)^{-1} = \left(1 + \frac{16}{4}\right)^{-1} = \boxed{0.2}$$

$$\mu_n = \tau_n^2 \left(\frac{\mu_0}{\tau_0^2} + \frac{n\bar{x}}{\sigma^2}\right) = 0.2\left(4 + \frac{16(5.4)}{4}\right)$$

$$= \boxed{5.12}$$

$$\mu \mid x \sim N(5.12,\ 0.2)$$

$$\bar{x}_p \mid x \sim N\left(5.12,\ \frac{0.2}{4}\right) = \bar{x}_4 \mid x \sim N(5.12, 4.2)$$

(e) $L \sim N(0, 1000^2)$

$\mu_0 = 0, \quad \tau_0^2 = 1,000,000$

$$\tau_n^2 = \left(\frac{1}{1000000} + \frac{16}{4} \right)^{-1} \approx \boxed{0.25}$$

$$\mu_n = \frac{0 + \frac{16(5.4)}{4}}{4.000001} \approx \boxed{5.4}$$

$P(x \sim N(5.40, 0.25))$

$\chi_P(x \sim N(5.40, 0.25))$

the variance of 0.25, this is small since data precision is 1/4 = .4 v.s. error precision of 1. our data with 0.4 60% of the weight while the likelihood error only has precision

2.

(a)

$$P(10 \le \mu \le 50) \approx 0.95$$

$$\mu_0 = \frac{10+50}{2} = \boxed{30} \qquad 30 - 2\tau_0 = 10$$

$$2\tau_0 = 2\tau_0^2$$

$$\tau_0^2 = 10$$

$$\mu \sim N(30, 100)$$

(6)

$$\pi(w) \propto 1 \quad , \quad -\infty < w < \infty$$

This improper because positive constant

can normalize this density to integrate

to 1

$$\pi(\mu|x) \propto L(\mu|x)\,\pi(\mu) \propto$$

$$\propto \exp\left[\frac{-1}{2\sigma^2}\sum^{n}\underbrace{(x_i-\mu)^2}\right]$$

$$\hookrightarrow \sum(x_i-\bar{x})^2 + n(\mu-\bar{x})^2$$

$$\pi(\mu|x) \propto \exp\left[\frac{-n}{2\sigma^2}(\mu-\bar{x})^2\right]$$

$$\boxed{\mu|x \sim N\left(\bar{x}, \frac{\sigma^2}{n}\right)}$$

the normalizing integral is finite for

$n \geq 1$ so the posterior is proper

only when the prior is Improper

$$\int_{-\infty}^{\infty} \exp\left[-\frac{n}{2\sigma^2} (\mu - \bar{x})^2 \right] d\mu = \sqrt{\frac{2\pi\sigma^2}{n}} < \infty$$

(c) $\sigma^2 \sim Unif(0,10)$

$$\tilde{\sigma} = \frac{1}{\sigma^2}$$

$$\pi_v(v) = \frac{1}{10}$$

$$v = \frac{1}{\lambda}, \quad \left|\frac{dv}{d\lambda}\right| = \frac{1}{\lambda^2}$$

$0 < v < 10 \Rightarrow$

$\lambda > \frac{1}{10}$

$$\pi_\lambda(\lambda) = \begin{cases} \frac{1}{10\lambda^2}, & \lambda > \frac{1}{10} \\ 0, & \text{otherwise} \end{cases}$$

$$\pi_\lambda(\lambda) = \pi_v(1/\lambda)\left|\frac{dv}{d\lambda}\right| = \frac{1}{10\lambda^2}$$

(1) $\pi(\tilde{\sigma}) \propto 1, \quad \tilde{\sigma} > 0$

$\nu = \sigma^2, \quad \lambda = \frac{1}{\nu}, \quad \left|\frac{d\lambda}{d\nu}\right| = \frac{1}{\nu^2}$

$\pi_\nu(\nu) = \pi_\lambda(1/\nu)\left|\frac{d\lambda}{d\nu}\right| \propto \frac{1}{\nu^2}, \quad \nu > 0$

$\pi(\sigma^2) \propto \frac{1}{(\sigma^2)^2}, \quad \sigma^2 > 0$

Since σ^2 is only constant, it's not =
to a float error or variance.

A error that is float in 1 parameterization
generally becomes fuzzing after a nonlinear
transformation. Since the Jacobian factor -
the parameterization used is what
we use to call a float error "non-informative

(c)

$$\pi_J(\lambda) \propto \frac{1}{\lambda}, \quad \lambda > 0$$

$$V = \sigma^2 = 1/\lambda$$

$$\pi_J(v) = \pi_J(1/v) \left| \frac{d\lambda}{dv} \right| \propto \frac{1}{1/v} \cdot \frac{1}{v^2}$$

$$= v \cdot \frac{1}{v^2} = \frac{1}{v}$$

$$\boxed{\pi_J(\sigma^2) \propto \frac{1}{\sigma^2}, \quad \sigma^2 > 0}$$

transforming features prior using the

Jacobian gives the same prior measure

as deriving Jeffreys prior directly in

the new parameterization

3.

$$x_1, \ldots, x_n \mid \sigma^2 \sim N(\mu, \sigma^2)$$

$$\tilde{\sigma} = \frac{1}{\sigma^2} \qquad SS_\mu = \sum (x_i - \mu)^2$$

(a)

$$f(x|\tilde{\sigma}) \propto \tilde{\sigma}^{n/2} \exp\left(-\frac{\tilde{\sigma}}{2} SS_\mu\right)$$

$$f(x|\sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left[-\frac{1}{2\sigma^2}\sum(x_i-\mu)^2\right]$$

$$\sigma^2 = 1/\eta \qquad SS_\mu = \sum(x_i-\mu)^2$$

$$f(x|\eta) = (2\pi)^{-n/2}\,\eta^{n/2} \exp\left[-\frac{\eta}{2} SS_\mu\right]$$

$$\boxed{f(x|\eta) \propto \eta^{n/2} \exp\left[-\frac{\eta}{2} SS_\mu\right]}$$

(4)

$$\lambda \sim \text{Gamma}(a, b)$$

$$\pi(\lambda) \propto \lambda^{a-1} e^{-b\lambda}$$

$$\pi(\lambda|x) \propto \lambda^{n/2} e^{-\lambda SS_n/2} \lambda^{a-1} e^{-b\lambda}$$

$$= \lambda^{a+n/2-1} \exp\left[-\left(b_0 + \frac{SS_n}{2}\right)\lambda\right]$$

$$a_n = a + \frac{n}{2}$$

$$b_n = b + \frac{SS_n}{2}$$

(c)

$$\pi_\tau(\tilde{\sigma}^2) \propto \frac{1}{\tilde{\sigma}^2}$$

$$\pi_\tau(\tau \mid x) \propto \tau^{n/2} e^{-\tau SS_\mu/2} \tau^{-1}$$

$$= \tau^{n/2-1} e^{-SS_\mu \tau/2}$$

$$\boxed{\tau \mid x \sim \text{Gamma}\left(\frac{n}{2}, \frac{SS_\mu}{2}\right)}$$

Posterior proper when $n > 0$ and $SS_x > 0$

its limiting case as $\alpha_0 \to 0^+$ and $\beta_0 \to 0^+$

the prior Kernel approaches π^{-1} and

the posterior learning approach

$\frac{n}{2}$ and $\frac{SS_x}{2}$, The scale parameter is

Inverse and Gamma$(2, 0)$ is not a

proper dist

6)

$n = 10 \qquad \mu = 5 \qquad SS_\mu = 18$

$\frac{n}{2} = 5, \qquad \frac{SS_\mu}{2} \to 9$

$a_n = 2 + 5 = 7$ $\qquad$ $\lambda | x \sim gamma(7, 10)$

$b_n = 1 + 0 = 10$ $\qquad$ $E[\lambda | x] = \frac{7}{10} = 0.7$

generic prior: $\lambda | x \sim gamma(5, 0)$ $\to$ $E(\lambda | x) = \frac{5}{9}$

(c) $\tau \mid x \sim \text{gamma}(a_n, b_n)$

$$x_f \mid \tau \sim N\left(\mu, \frac{1}{\tau}\right)$$

$$p(x_f \mid x) \propto \left[1 + \frac{(x_f - \mu)^2}{2b_n}\right]^{-(a_n + 1/2)}$$

$$\left[1 + \frac{(x_f - m)^2}{V c^2}\right]^{-(\nu+1)/2}$$

$$\nu = 2\alpha_n, \quad m = \mu, \quad s = \sqrt{\frac{\alpha_n}{\beta_n}}$$

$$\chi_t \mid \chi \sim t(\text{location} = s,\ \text{scale} = \sqrt{\frac{\alpha_n}{\beta_n}})$$

degrees of freedom $= 16$

4.

$$x_1, \ldots, x_n \mid \mu, \sigma \sim N(\mu, \sigma^2)$$

$$\mu \mid \sigma^2 \sim N\left(\mu_0, \frac{\sigma^2}{\varphi_0}\right)$$

$$\sigma^2 \sim IG\left(\frac{\nu_0}{2}, \frac{SS_0}{2}\right)$$

$$SS = \sum (x_i - \bar{x})^2$$

(a)

$$\rho_n = \rho_0 + n$$

$$\mu_n = \frac{\rho_0 \mu_0 + n\bar{x}}{\rho_0 + n}$$

$$\nu_n = \nu_0 + n$$

$$L(\mu, \sigma^2 | x) \propto (\sigma^2)^{-n/2} \exp\left[\frac{-SS + n(\mu - \bar{x})^2}{2\sigma^2} \right]$$

$$\sum (x_i - \mu)^2 = SS + n(\mu - \bar{x})^2$$

$$\pi(\mu, \sigma^2 | x) \propto (\sigma^2)^{-(\frac{\nu_0 + n}{2} + 3/2)} \exp\left[-\frac{SS_0 + SS + n(\mu - \bar{x})^2 + \rho_0(\mu - \mu_0)^2}{2\sigma^2} \right]$$

$$n(\mu - \bar{x})^2 + \rho_0(\mu - \mu_0)^2 = (\rho_0 + n)(\mu - \mu_n)^2 + \frac{\rho_0 n}{\rho_0 + n}(\bar{x} - \mu_0)^2$$

$$\text{posterior} \propto (\sigma^2)^{-1/2} e^{-\rho_0(\mu-\tau_0)^2/(2\sigma^2)} (\sigma^2)^{-(\nu_0/2+1)} e^{-SS_0/2\sigma^2}$$

$$\mu|\sigma^2, x \sim N\left(\mu_0, \frac{\sigma^2}{\rho_0}\right)$$

$$\sigma^2|x \sim IG\left(\frac{\nu_0}{n}, \frac{SS_0}{2}\right)$$

(6) $\mu_n = \frac{\rho_0}{\rho_0 + n} \nu_0 + \frac{n}{\rho_0 + n} \bar{x}$

the ρ_0 is the weight and ν_0 and prior mean and same mean.

It acts Acc for estimating μ so gives more weight to prior mean

(C)

$$SS_h = SS_0 + SS + \frac{\rho_{0h}}{\rho_0 + h}(\bar{x} - \mu_0)^2$$

SS_0 is the prior contribution to our sum of squares parameter while SS is $\sum (x_i - \bar{x})^2$ which measures the variation of the observations around their sample mean and $\frac{\rho_0 h}{\rho_0 + h}(\bar{x} - \mu_0)^2$ is the disagreement prior mean vs. sample mean

the difference between prior and sample

mean is squared which increases our

posterior variance scale

(d)

$n_0 = 4$, $K_0 = 10$, $V_0 = 6$, $SS_0 = 30$

$n = 8$, $\bar{x} = 12$, $SS = 42$

$$\varphi_n = n + \delta = 12$$

$$K_n = \frac{9(10) + 8(14)}{12} = \frac{34}{3} \simeq 11.33$$

$$V_n = 6 + 8 = 14$$

$$SS_2 = 30 + 42 + \frac{4(8)}{12}(12-10)^2$$

$$= \frac{245}{3}$$

$$\bar{x}_1 \mid \sigma^2 \sim N\left(\mu_1, \frac{\sigma^2}{12}\right)$$

$$\sigma^2 / x \sim IG\left(7, \frac{124}{3}\right)$$

(c) $\mu | x \sim t_{\nu_h}\left(\mu_h, \sqrt{\frac{SS_h}{\rho_h \nu_h}}\right)$

$\sqrt{\frac{SS_h}{\rho_h \nu_h}} = \sqrt{\frac{248}{3}} = \sqrt{\frac{31}{63}}$

$\mu | x \sim t_{(4)}\left(\text{location} = \frac{34}{3},\ \text{scale} = \sqrt{\frac{31}{63}}\right)$

5. (a) Code in R file

$h = 10, \quad \bar{x} = 10.07, \quad s_3 = 2.001$

$p_h = 13, \quad \phi_n = \frac{3(9) + 10(10.09)}{13} = \frac{127}{13}$

$\nu_n = 14$

$$SS_h = 8 + 7.001 + \frac{30}{13}(10.07 - 9)^2$$

$$= 17.643$$

$$\sigma^2 | x \sim IG(7, 8.82)$$

$$\mu | \sigma^2, x \sim N(9.82, \frac{\sigma^2}{13})$$

(b) Code written

(c) $E[\mu|x] = 9.822$

$P(\mu > 10|x) = 0.2866$

28.7% is our posterior probability

that $\mu > 10$

(d)

$$E[\sigma^2 / x] \simeq 1.4628$$

$$(0.677,\ 3.1288)$$

(6) $\mu_0 = 0, \quad \rho_0 = 0.01,$

$\nu_0 = 0.01, \quad SS_0 = 0.01$

$\rho_n = \rho 0.01, \quad \mu_n = 10.0599, \quad \nu_n = 10.01$

$$SS_n = 0.01 + 2.0001 + \frac{0.01(10)}{10.01}(10.02-0)^2$$

$$= \boxed{8.0240}$$

$$\sigma^2 \mid x \sim IG(5.005,\ 4.012)$$

$$\mu \mid \sigma^2, x \sim N\left(10.0590,\ \frac{\sigma^2}{50.01}\right)$$